

Unit 3 – Cyber Crime with Mobile and Wireless Devices

Complete 10–15 Marks Exam Notes

Below are **exam-oriented answers** for every topic in the syllabus. You can write these directly in a **10–15 mark question**. I have kept the language simple and added **diagrams, examples, advantages/importance, attacks and prevention points** wherever useful.

1. Cyber Crime with Mobile and Wireless Devices

Introduction

A **mobile device** is a portable electronic device such as a smartphone, tablet, smartwatch, etc. that can connect to the Internet or wireless networks.

Cyber crime with mobile and wireless devices refers to illegal activities performed using mobile phones, wireless networks or targeting mobile devices to steal information, money, identity or services.

Examples:

- Mobile banking fraud
- UPI/payment fraud
- SIM swapping
- Phishing through SMS
- Malicious mobile applications
- Identity theft
- Wi-Fi attacks
- Bluetooth attacks
- Data theft
- Mobile ransomware

Basic diagram

MOBILE / WIRELESS CYBER CRIME

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+-----+-----+		
Mobile	Wireless	Online
Device	Network	Services
Malware	Wi-Fi Attack	Banking Fraud
Spyware	MITM	Phishing
Ransomware	Evil Twin	Identity Theft
Data Theft	Sniffing	Payment Fraud

Common Mobile Cyber Crimes

1. Mobile Malware

Malicious software installed on a mobile device to steal information or damage the device.

Examples:

- Spyware
- Trojan
- Ransomware
- Banking malware

2. Phishing and Smishing

Phishing uses fake websites/emails.

Smishing uses fraudulent SMS messages.

Example:

"Your bank account will be blocked. Click here to verify."

The link may lead to a fake banking website.

3. SIM Swap Fraud

An attacker obtains control of the victim's mobile number by transferring the number to another SIM.

The attacker may then receive:

- OTPs
- Banking messages
- Password-reset codes

4. Mobile Payment Fraud

Attackers may target:

- UPI
- Mobile banking
- Digital wallets
- Credit/debit cards

5. Identity Theft

Personal information such as:

- Name
- Mobile number
- Aadhaar-related information
- Email
- Passwords

may be stolen and misused.

6. Bluetooth Attacks

Attackers can attempt to exploit insecure Bluetooth connections to access or transfer information.

7. Wi-Fi Attacks

Attackers may create fake Wi-Fi networks or intercept poorly protected communications.

Impact

Mobile cyber crimes can result in:

1. Financial loss
 2. Identity theft
 3. Privacy violation
 4. Loss of confidential data
 5. Reputation damage
 6. Unauthorized transactions
 7. Business losses
 8. Emotional stress
 9. Legal consequences
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Prevention

- Use strong passwords/PINs.
- Enable biometric authentication.
- Install applications only from trusted sources.
- Keep OS and applications updated.
- Avoid unknown Wi-Fi networks.
- Do not click suspicious SMS links.
- Enable device encryption.
- Use multi-factor authentication.
- Lock the device automatically.
- Enable remote tracking/wiping.
- Review application permissions.

Conclusion

Mobile devices have become an important part of communication, banking and business. Therefore, securing mobile devices and wireless communication is essential to prevent financial, privacy and identity-related cyber crimes.

2. Introduction and Proliferation of Mobile and Wireless Devices

Introduction

The rapid growth in the use of smartphones, tablets, laptops, smartwatches and IoT devices is called the **proliferation of mobile and wireless devices**.

Wireless technology allows devices to communicate without physical cables.

Examples:

- Wi-Fi
- Bluetooth
- NFC
- 4G/5G
- GPS
- Mobile networks

Evolution

Basic Mobile



Feature Phone



Smartphone



4G / 5G Smartphone



IoT + Wearables + Smart Devices

Reasons for Rapid Growth

1. Internet availability

Affordable mobile Internet has increased smartphone usage.

2. Affordable smartphones

Smartphones are now available in different price ranges.

3. Mobile applications

Apps are available for:

- Banking
- Education
- Shopping
- Entertainment
- Healthcare
- Transportation

4. Digital payments

UPI, mobile banking and wallets have increased dependence on smartphones.

5. Remote work and education

Mobile and wireless technologies allow users to work and study remotely.

6. IoT growth

Devices such as:

- Smart TVs
- Smart watches
- Smart cameras
- Smart speakers

use wireless connectivity.

Advantages

- Easy communication
- Internet access anywhere
- Mobile banking
- Online education

- Remote working
 - Location-based services
 - Digital payments
 - Entertainment
 - Emergency communication
-

Security Problems

The increased number of connected devices also creates more opportunities for attackers.

Common risks:

- Malware
- Data theft
- Phishing
- Wi-Fi attacks
- Identity theft
- Password attacks
- Unauthorized access
- Privacy violations

Important concept

More connected devices → larger attack surface → greater security requirements

Conclusion

The proliferation of mobile and wireless devices has transformed modern society. However, increased connectivity also increases cyber-security risks, making authentication, encryption, secure applications and user awareness important.

3. Trends in Mobility

Meaning

Mobility trends describe the major changes in the way people use mobile devices, wireless networks and mobile applications.

Major Trends

1. Smartphone Dominance

Smartphones are increasingly being used for:

- Communication
- Banking
- Shopping
- Education
- Entertainment
- Work

2. 5G Technology

5G provides:

- Higher speed
- Lower latency
- Greater device connectivity
- Support for IoT

3. Mobile Cloud Computing

Mobile devices use cloud services to store and process data.

Mobile Device

|

| Internet

↓

Cloud Server

|

↓

Applications + Storage + Processing

4. Bring Your Own Device (BYOD)

Employees use their personal smartphones/laptops for organizational work.

Benefits:

- Flexibility
- Reduced hardware cost
- Increased productivity

Risks:

- Data leakage
- Malware
- Unauthorized access
- Loss of device

5. Internet of Things

More physical objects are becoming connected to the Internet.

Examples:

- Smartwatch
- Smart home
- Smart vehicle
- Smart camera

6. Mobile Payments

UPI, wallets and mobile banking are rapidly increasing.

7. Wearable Technology

Examples:

- Smartwatch
- Fitness tracker
- Smart glasses

These devices collect sensitive personal data.

8. Remote Working

Mobile devices allow employees to access organizational resources remotely.

9. AI-powered Mobile Applications

AI is increasingly used in:

- Voice assistants
- Image recognition
- Fraud detection
- Recommendation systems

Security Challenges of Mobility

Mobility



More Devices



More Connectivity



More Data



Larger Attack Surface



More Security Requirements

Conclusion

Modern mobility is moving toward **5G, cloud computing, IoT, mobile payments, BYOD and AI**. These technologies provide convenience but require strong security mechanisms.

4. Credit Card Fraud – Types

Definition

Credit card fraud is the unauthorized use of a credit card or credit-card information to obtain money, goods or services.

The attacker may steal:

- Card number
- CVV
- Expiry date
- PIN
- OTP
- Account credentials

Types of Credit Card Fraud

1. Lost or Stolen Card Fraud

The physical card is lost or stolen and used by another person.

2. Card-Not-Present Fraud

The physical card is not required.

It commonly occurs during:

- Online shopping
- Online subscriptions
- Telephone transactions

The attacker only needs stolen card information.

3. Phishing

Victim receives a fake email/SMS/website requesting card details.

Example:

Fake Bank Message



Fake Website



Card Details



Attacker

4. Skimming

A skimming device captures card information during a transaction.

It can target:

- ATMs
- Payment terminals

5. Identity Theft

An attacker obtains personal information and uses it to create or misuse financial accounts.

6. Account Takeover

The attacker gains access to the victim's online banking/card account.

7. Application Fraud

Fraudulent personal information may be used to apply for a new credit card.

8. Card Cloning

Stolen card information can potentially be used to create a fraudulent duplicate card.

5. Impact of Credit Card Fraud

Credit card fraud affects individuals, banks and businesses.

1. Financial Loss

Unauthorized transactions can cause direct monetary loss.

2. Identity Theft

Personal and financial information may be misused.

3. Credit Score Problems

Fraudulent transactions or accounts may negatively affect the victim's credit history.

4. Privacy Loss

Sensitive financial information may be exposed.

5. Business Loss

Merchants and financial institutions may face fraud-related costs.

6. Reputation Damage

Banks and businesses can lose customer trust.

7. Investigation Costs

Organizations may spend significant resources investigating fraudulent activities.

6. Strategies to Prevent Credit Card Fraud

1. Multi-Factor Authentication

Use more than one authentication factor.

Password

+

OTP / Biometric

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Access

2. Transaction Alerts

Enable SMS/email/app notifications for every transaction.

3. Secure Online Shopping

Check that the website uses HTTPS and is a legitimate website.

4. Never Share OTP

Banks generally advise customers not to share OTPs, PINs or CVVs with others.

5. Use Transaction Limits

Set suitable spending/transaction limits.

6. Fraud Detection Systems

Banks use machine-learning and rule-based systems to detect unusual transactions.

Example:

Normal Transaction



Customer buys locally



Normal

Unusual Transaction



Sudden high-value transaction



Risk Analysis



Possible Fraud Alert

7. Card Blocking

Immediately block the card if it is lost or stolen.

8. Virtual/Tokenized Cards

Tokenization replaces sensitive card information with a token.

9. Regular Monitoring

Check:

- Bank statements
- Card transactions
- Account activity

Conclusion

Credit card fraud can cause financial and privacy losses. Strong authentication, transaction monitoring, fraud detection and user awareness can significantly reduce the risk.

7. Security Challenges of Mobile Devices

Mobile devices have unique security challenges because they are portable, connected and contain large amounts of personal information.

Major Challenges

1. Device Theft or Loss

A stolen phone can expose:

- Photos
- Contacts
- Emails
- Applications
- Documents

2. Malware

Malicious apps may steal information or monitor activities.

3. Weak Authentication

Simple PINs or passwords can be guessed or compromised.

4. Insecure Applications

Poorly developed applications may contain vulnerabilities.

5. Unsecured Wi-Fi

Public Wi-Fi can create risks if communications are not properly protected.

6. Data Leakage

Applications may access or transmit more information than necessary.

7. Outdated Software

Old operating systems may contain known vulnerabilities.

8. Phishing and Social Engineering

Users can be tricked through SMS, email, calls or fake applications.

9. Bluetooth/NFC Risks

Wireless communication technologies can introduce additional attack surfaces.

10. Privacy Issues

Applications may collect:

- Location
 - Contacts
 - Camera data
 - Microphone data
 - Device information
-

Security Measures

- Strong authentication
 - Encryption
 - Secure application development
 - Regular updates
 - Antivirus/mobile security tools
 - Secure Wi-Fi
 - Permission management
 - Remote wipe
 - Employee awareness
-

8. Mitigation Strategies for Mobile Security

Mitigation means reducing the probability or impact of a security threat.

1. Strong Authentication

Use:

- Strong password
- PIN
- Fingerprint
- Face authentication
- MFA

2. Encryption

Encrypt data stored on the device and data transmitted through networks.

3. Regular Updates

Keep:

- Operating system
- Applications
- Security software

updated.

4. Application Control

Install applications only from trusted sources.

5. Permission Management

Review whether an application really requires:

- Camera
- Microphone
- Location
- Contacts
- Storage

6. Remote Lock/Wipe

If the phone is lost, it can be remotely locked or erased.

7. Secure Network

Avoid transmitting sensitive information over unsecured public networks.

8. Backup

Important information should be backed up securely.

9. Security Awareness

Users should be trained to identify:

- Phishing
- Fake applications
- Fraudulent calls
- Suspicious links

9. Registry Setting for Mobile Devices

Meaning

Registry/settings configuration refers to configuring operating-system and security settings that control how a mobile device operates and how applications access device resources.

In practical mobile security, this includes controlling:

- Security settings
- Application permissions
- Network settings
- Authentication
- Encryption
- Device restrictions
- Storage
- Updates

In exams, write this topic as **secure configuration and registry/settings management of mobile devices**.

Important Security Settings

1. Screen Lock

Use:

- PIN
- Password
- Pattern
- Biometrics

2. Automatic Lock

The device should automatically lock after a period of inactivity.

3. Encryption

Enable device/storage encryption.

4. Application Permissions

Restrict unnecessary permissions.

5. Unknown Sources

Prevent installation of applications from untrusted sources.

6. Developer Options

Organizations may restrict unnecessary developer/debugging features.

7. Network Configuration

Use secure Wi-Fi and avoid unnecessary network connections.

8. Bluetooth

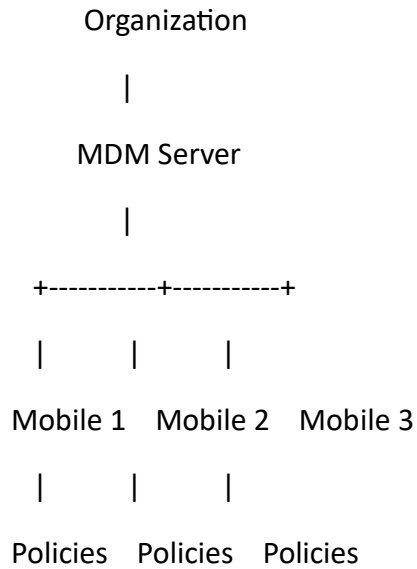
Bluetooth should be disabled when not required and configured securely.

9. Automatic Updates

Enable security and OS updates.

10. Remote Management

Organizations can use Mobile Device Management (MDM) to enforce security policies.



Benefits

- Centralized security
- Policy enforcement
- Remote device management
- Remote wiping
- Application control
- Better compliance

10. Authentication Service Security

Definition

Authentication is the process of verifying whether a user is actually who they claim to be.

Example:

User



Username + Password



Authentication Server



Verification



Access Granted / Denied

Authentication Factors

1. Something You Know

Examples:

- Password
- PIN
- Security answer

2. Something You Have

Examples:

- Smartphone
- Security token
- Smart card

3. Something You Are

Examples:

- Fingerprint
- Face
- Iris

4. Somewhere You Are

Location-based authentication.

5. Something You Do

Behavioral characteristics such as typing patterns.

Multi-Factor Authentication

MFA combines two or more authentication factors.

Example:

Password

+

OTP

+

Fingerprint

↓

Authenticated User

Authentication Security Threats

- Password guessing
- Credential theft
- Phishing
- Credential stuffing
- Brute-force attacks
- Session hijacking
- SIM-based OTP interception

Security Measures

Strong Password Policy

Passwords should be difficult to guess.

MFA

Use multiple authentication factors.

Account Lockout/Rate Limiting

Repeated failed attempts should trigger restrictions.

Secure Password Storage

Passwords should not be stored as plain text.

Session Security

Sessions should be protected against unauthorized access.

Biometric Authentication

Biometric authentication provides an additional factor, especially when combined with a secure device credential.

Conclusion

Authentication is a critical component of mobile security because it prevents unauthorized users from accessing devices, applications and data.

11. Attacks on Mobile Devices

Mobile devices can be attacked through applications, networks, social engineering and physical access.

1. Malware Attack

Malicious applications can:

- Steal data
- Monitor activity
- Encrypt files
- Perform unauthorized actions

Types

- Trojan
- Spyware
- Ransomware
- Adware

2. Phishing/Smishing

Fake messages attempt to convince users to reveal sensitive information.

Fake SMS



Malicious Link



Fake Website



Credentials Stolen

3. Man-in-the-Middle Attack

An attacker attempts to intercept communication between two parties.

User $\leftrightarrow$ Attacker $\leftrightarrow$ Server

Proper encryption and certificate validation help mitigate this risk.

4. Evil Twin Attack

An attacker creates a fake Wi-Fi network that resembles a legitimate network.

Example:

Real Wi-Fi: College_WiFi

Fake Wi-Fi: College_WiFi



Attacker

Users may accidentally connect to the fake network.

5. SIM Swap

The attacker obtains control of the victim's phone number.

Potential consequences include:

- OTP interception
- Account recovery abuse

- Unauthorized transactions
-

6. Bluetooth Attacks

Poorly configured Bluetooth can expose devices to unauthorized communication.

7. Malicious Application

A seemingly useful application may contain malicious functionality.

8. Ransomware

Attackers may encrypt or lock data and demand payment.

9. Spyware

Spyware secretly monitors activities or collects information.

10. Physical Attacks

A stolen or unattended phone can provide access if it is poorly protected.

Prevention

- Use official application stores.
- Keep software updated.
- Use strong authentication.
- Encrypt sensitive data.
- Avoid suspicious links.
- Use secure Wi-Fi.
- Review permissions.
- Enable remote tracking/wiping.

- Do not root/jailbreak devices unnecessarily.
-

12. Organizational Measures for Handling Mobile Security

Organizations must create policies and technical controls to protect mobile devices used for business.

The major measures are:

Mobile Security

- |
 - +-- Mobile Device Management
 - |
 - +-- Authentication
 - |
 - +-- Application Security
 - |
 - +-- Data Protection
 - |
 - +-- Network Security
 - |
 - +-- Employee Training
-

13. Implementing Strong Mobile Device Management Policy

Definition

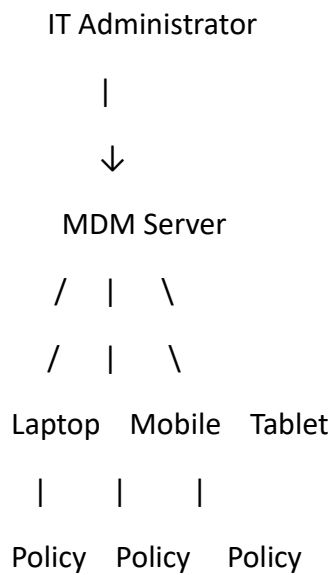
Mobile Device Management (MDM) is a system that allows an organization to centrally manage and secure mobile devices.

MDM can control:

- Device configuration
- Applications

- Security policies
- Password requirements
- Encryption
- Network access
- Remote lock
- Remote wipe

Architecture



Important MDM Policies

1. Password Policy

Require strong device passwords/PINs.

2. Encryption Policy

Require encryption for organizational data.

3. Application Policy

Block unauthorized applications.

4. Remote Wipe

Allow administrators to erase organizational data from a lost device.

5. Device Compliance

Devices must satisfy security requirements before accessing company resources.

6. BYOD Policy

Define security requirements for employee-owned devices.

14. Enforcing Strong Authentication and Access Controls

Authentication

Determines:

Who are you?

Authorization

Determines:

What are you allowed to access?

Example

Employee



Login



Authentication



Identity Verified



Authorization



Access only permitted resources

Access Control Principles

Least Privilege

Users should receive only the permissions they need.

Role-Based Access Control

Permissions are assigned according to job role.

Example:

Role	Access
Student	Own records
Faculty	Student academic records
Administrator	Administrative data

MFA

Use multiple authentication factors.

Session Management

Secure sessions and terminate inactive sessions.

15. Securing Mobile Applications and Data

Mobile Application Security

Mobile applications may process highly sensitive data.

Examples:

- Banking applications
- Healthcare applications
- Payment applications
- Educational applications

Security Measures

1. Secure Coding

Developers should follow secure programming practices.

2. Input Validation

Applications should validate input to reduce security vulnerabilities.

3. Encryption

Sensitive data should be encrypted.

4. Secure API

APIs should use authentication and authorization.

5. Secure Storage

Do not store sensitive information insecurely.

6. Permission Management

Applications should request only required permissions.

7. Regular Security Testing

Perform:

- Vulnerability assessment
- Security testing
- Code review

8. Secure Authentication

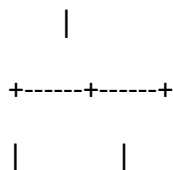
Use:

- MFA
- Secure tokens
- Strong session management

Data Security

Data should be protected:

MOBILE DATA



Data at Rest Data in Transit

| |
Encryption TLS/HTTPS

Data at Rest

Data stored on the phone.

Data in Transit

Data travelling between mobile device and server.

Both require appropriate security controls.

16. Protecting Wireless Network Connections

Definition

Wireless network security means protecting Wi-Fi and other wireless communications from unauthorized access, interception and manipulation.

Common Wireless Threats

1. Eavesdropping

An attacker attempts to observe wireless communication.

2. Evil Twin

Fake Wi-Fi network pretending to be legitimate.

3. Man-in-the-Middle

Attacker attempts to intercept communication.

4. Unauthorized Access

An unauthorized user attempts to connect to the network.

5. Denial of Service

An attacker attempts to make the wireless service unavailable.

Security Measures

1. Strong Wi-Fi Encryption

Use modern secure wireless security protocols such as WPA2/WPA3 where supported.

2. Strong Password

Use a strong wireless network password.

3. Change Default Credentials

Router default administrative credentials should be changed.

4. Firmware Updates

Keep routers/access points updated.

5. Network Segmentation

Separate sensitive systems from guest devices.

Internet

|

Router

|

+---- Corporate Network

|

+---- Guest Network

|

+---- IoT Network

6. VPN

A VPN can provide an additional protected communication channel when appropriately configured.

7. Disable Unnecessary Features

Disable unnecessary wireless services.

8. Monitor Network

Monitor for suspicious devices and unusual network activity.

17. Educating Employees About Mobile Security

Employees are often an important part of an organization's security system.

Even strong technical security can be weakened by unsafe user behavior.

Employee Training Should Cover

1. Phishing Awareness

Employees should learn how to recognize suspicious:

- Emails
- SMS
- Links
- Websites
- QR codes

2. Password Security

Employees should:

- Use strong passwords.
- Avoid password reuse.
- Use MFA.

3. Device Security

Employees should:

- Lock their devices.
- Avoid leaving devices unattended.
- Report lost devices immediately.

4. Safe Application Usage

Employees should install applications only from trusted sources.

5. Public Wi-Fi Awareness

Employees should avoid using insecure public networks for sensitive activities.

6. Data Protection

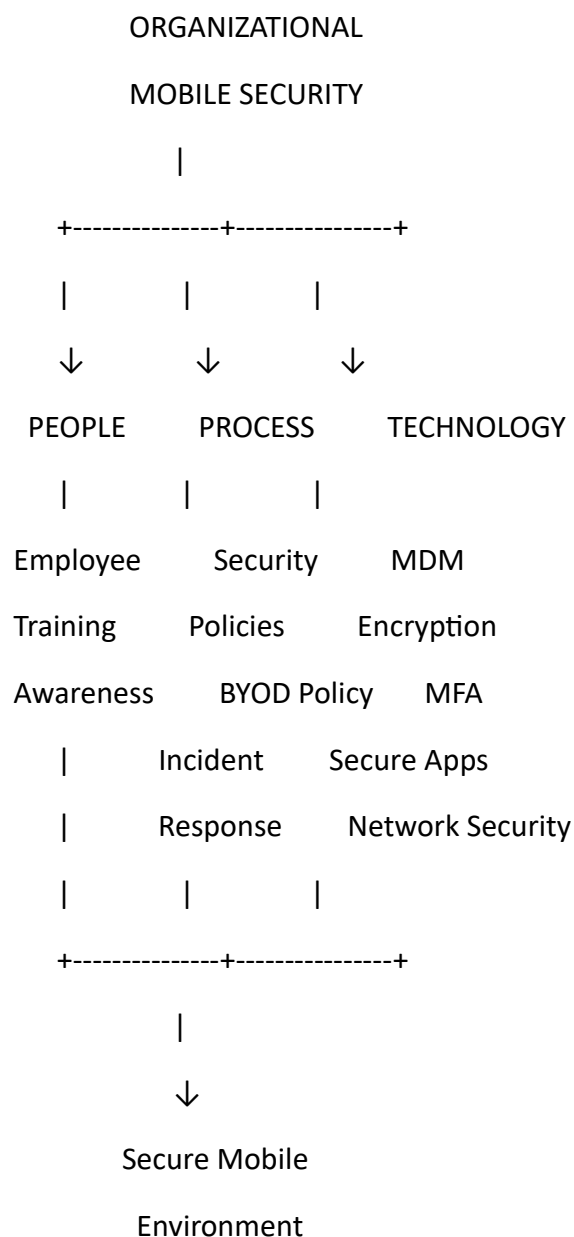
Employees should not share confidential organizational information through unauthorized applications or services.

7. Social Engineering Awareness

Employees should understand manipulation techniques used by attackers.

18. Complete Organizational Mobile Security Framework

This is a **very important diagram for a 10–15 mark answer**.



19. Overall Mobile Security Strategy

An organization should follow a **defense-in-depth approach**.

Layer 1 – Device Security

- Screen lock
- Encryption
- Updates
- Antivirus/security tools

Layer 2 – Application Security

- Secure applications
- Permission management
- Secure APIs

Layer 3 – Network Security

- WPA2/WPA3
- VPN
- Network segmentation
- Monitoring

Layer 4 – Authentication

- Strong passwords
- MFA
- Biometrics

Layer 5 – Data Security

- Encryption
- Secure storage
- Backup
- Data loss prevention

Layer 6 – Human Security

- Employee training
- Security awareness
- Incident reporting

SECURITY

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| | |

Device Network Data

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Authentication

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Employee Awareness

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Incident Response

★ Quick Revision Table

Topic	Main Points to Remember
Cyber crime with mobile devices	Malware, phishing, SIM swap, payment fraud
Proliferation	Rapid growth of smartphones, IoT, wireless
Mobility trends	5G, cloud, BYOD, IoT, mobile payment
Credit card fraud	Lost card, CNP, phishing, skimming, identity theft
Impact	Financial loss, privacy loss, reputation

Topic	Main Points to Remember
Prevention	MFA, alerts, monitoring, tokenization
Mobile security challenges	Malware, theft, Wi-Fi, apps, privacy
Mitigation	Encryption, updates, authentication, backup
Registry/settings	Secure device configuration, permissions, MDM
Authentication	Verify identity
Authentication factors	Know, have, are, location/behavior
Mobile attacks	Malware, phishing, MITM, evil twin, SIM swap
MDM	Centralized mobile device management
Access control	Authentication + authorization + least privilege
Mobile app security	Secure coding, encryption, API security
Wireless security	WPA2/WPA3, VPN, segmentation
Employee security	Training, phishing awareness, safe device usage



How to Write a 10–15 Mark Answer in Exam

For **almost every topic in this unit**, use this structure:

1. Definition – 1–2 marks

Write a clear definition.

2. Introduction – 1 mark

Explain why the topic is important.

3. Diagram – 1–2 marks

Draw a simple flowchart/architecture.

4. Main explanation – 5–6 marks

Write **6–8 subheadings** with 2–4 lines each.

5. Examples – 1 mark

Give 2–3 practical examples.

6. Prevention/Security Measures – 2–3 marks

Write relevant security controls.

7. Conclusion – 1 mark

End with 2–3 lines summarizing the topic.

Example answer flow

DEFINITION



INTRODUCTION



DIAGRAM



TYPES / COMPONENTS



WORKING / EXPLANATION



EXAMPLES



SECURITY / PREVENTION



ADVANTAGES / IMPACT



CONCLUSION

Exam tip: For a 10–15 mark question, don't write one huge paragraph. Use **headings + bullet points + diagrams + examples**. This makes the answer easier for the examiner to evaluate and helps you cover more points.

